

TRIGGERS AND STORED PROCEDURES

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1. CREATE DATABASE

The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying two SQL statements: `CREATE DATABASE BankDB;` and `USE BankDB;`. The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message
✓ 1	12:31:16	CREATE DATABASE BankDB	1 row(s) affected
✓ 2	12:31:16	USE BankDB	0 row(s) affected

2. CREATE CUSTOMER TABLE

The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying the following SQL statements:

```
1 • CREATE DATABASE BankDB;
2 • USE BankDB;
3
4 • CREATE TABLE Customer (
5     Customer_ID INT PRIMARY KEY,
6     Customer_Name VARCHAR(100) NOT NULL,
7     Phone VARCHAR(15),
8     Email VARCHAR(100),
9     City VARCHAR(50)
10 );
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message
✓ 1	12:31:16	CREATE DATABASE BankDB	1 row(s) affected
✓ 2	12:31:16	USE BankDB	0 row(s) affected
✗ 3	12:34:08	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; database .
✓ 4	12:34:08	USE BankDB	0 row(s) affected
✓ 5	12:34:09	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected

3. CREATE ACCOUNT TABLE

The screenshot shows the SQL Developer interface with the 'TRIGGERS_AND_STORED_P...' window open. The SQL editor contains the following code:

```
5 Customer_ID INT PRIMARY KEY,  
6 Customer_Name VARCHAR(100) NOT NULL,  
7 Phone VARCHAR(15),  
8 Email VARCHAR(100),  
9 City VARCHAR(50)  
10 );  
11  
12 CREATE TABLE Account (  
13 Account_No INT PRIMARY KEY,  
14 Customer_ID INT,  
15 Account_Type VARCHAR(20),  
16 Balance DECIMAL(12,2) DEFAULT 0,  
17 Branch VARCHAR(50),  
18  
19 FOREIGN KEY (Customer_ID)  
20 REFERENCES Customer(Customer_ID)  
21 );  
22
```

The Output window shows the following messages:

#	Time	Action	Message
3	12:34:08	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; databas...
4	12:34:08	USE BankDB	0 row(s) affected
5	12:34:09	CREATE TABLE Customer (Customer_ID INT PRIMAR...	0 row(s) affected
6	12:37:07	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; databas...
7	12:37:07	USE BankDB	0 row(s) affected
8	12:37:07	CREATE TABLE Customer (Customer_ID INT PRIMAR...	Error Code: 1050. Table 'customer' already exists
9	12:37:07	CREATE TABLE Account (Account_No INT PRIMARY...	0 row(s) affected

4. CREATE TRANSACTION TABLE

The screenshot shows the SQL Developer interface with the 'TRIGGERS_AND_STORED_P...' window open. The SQL editor contains the following code:

```
15 Account_Type VARCHAR(20),  
16 Balance DECIMAL(12,2) DEFAULT 0,  
17 Branch VARCHAR(50),  
18  
19 FOREIGN KEY (Customer_ID)  
20 REFERENCES Customer(Customer_ID)  
21 );  
22  
23 CREATE TABLE Bank_Transaction (  
24 Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,  
25 Account_No INT,  
26 Transaction_Type VARCHAR(20),  
27 Amount DECIMAL(12,2),  
28 Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,  
29  
30 FOREIGN KEY (Account_No)  
31 REFERENCES Account(Account_No)  
32 );
```

The Output window shows the following messages:

#	Time	Action	Message
3	13:37:45	CREATE TABLE Customer (Customer_ID INT PRIMAR...	Error Code: 1050. Table 'customer' already exists
4	13:37:45	CREATE TABLE Account (Account_No INT PRIMARY...	Error Code: 1050. Table 'account' already exists
5	13:39:08	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; databas...
6	13:39:08	USE BankDB	0 row(s) affected
7	13:39:08	CREATE TABLE Customer (Customer_ID INT PRIMAR...	Error Code: 1050. Table 'customer' already exists
8	13:39:08	CREATE TABLE Account (Account_No INT PRIMARY...	Error Code: 1050. Table 'account' already exists
9	13:39:08	CREATE TABLE Bank_Transaction (Transaction_ID IN...	0 row(s) affected

5. CREATE LOAN TABLE

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Filter objects: bank_db, bank_managemen..., bankdb, Tables, Views, Stored Proc..., Functions, bookflow_db, emr_portal, hospital_db, joins_db, practice_db, sakila, student_db, student_info, student_marks1..., student_marks_c..., students_db, sys, world.

SQL Editor: TRIGGERS_AND_STORED_P...

```

26 Transaction_Type VARCHAR(20),
27 Amount DECIMAL(12,2),
28 Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
29
30 FOREIGN KEY (Account_No)
31 REFERENCES Account(Account_No)
32 );
33
34 CREATE TABLE Loan (
35 Loan_ID INT PRIMARY KEY,
36 Customer_ID INT,
37 Loan_Type VARCHAR(30),
38 Loan_Amount DECIMAL(12,2),
39 Interest_Rate DECIMAL(5,2),
40
41 FOREIGN KEY (Customer_ID)
42 REFERENCES Customer(Customer_ID)
43 );

```

Output: Action Output

#	Time	Action	Message
9	13:39:08	CREATE TABLE Bank_Transaction (Transaction_ID IN...	0 row(s) affected
10	13:40:04	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; databas...
11	13:40:04	USE BankDB	0 row(s) affected
12	13:40:04	CREATE TABLE Customer (Customer_ID INT PRIMAR...	Error Code: 1050. Table 'customer' already exists
13	13:40:04	CREATE TABLE Account (Account_No INT PRIMARY...	Error Code: 1050. Table 'account' already exists
14	13:40:04	CREATE TABLE Bank_Transaction (Transaction_ID IN...	Error Code: 1050. Table 'bank_transaction' already exists
15	13:40:04	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected

Object Info Session

6. INSERT CUSTOMER DATA

SQL Enterprise Manager - J.Simran-2520030166 - Warni...

Filter objects: bank_db, bank_managemen..., bankdb, Tables, Views, Stored Proc..., Functions, bookflow_db, emr_portal, hospital_db, joins_db, practice_db, sakila, student_db, student_info, student_marks1..., student_marks_c..., students_db, sys, world.

SQL Editor: TRIGGERS_AND_STORED_P...

```

34 Loan_ID INT PRIMARY KEY,
35 Customer_ID INT,
36 Loan_Type VARCHAR(30),
37 Loan_Amount DECIMAL(12,2),
38 Interest_Rate DECIMAL(5,2),
39
40 FOREIGN KEY (Customer_ID)
41 REFERENCES Customer(Customer_ID)
42 );
43
44 INSERT INTO Customer
45 (Customer_ID, Customer_Name, Phone, Email, City)
46 VALUES
47 (101, 'Ravi Kumar', '9876543210', 'ravi@gmail.com', 'Hyderabad'),
48 (102, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'),
49 (103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),
50 (104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
51 (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');

```

Output: Action Output

#	Time	Action	Message
10	13:40:04	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; databas...
11	13:40:04	USE BankDB	0 row(s) affected
12	13:40:04	CREATE TABLE Customer (Customer_ID INT PRIMAR...	Error Code: 1050. Table 'customer' already exists
13	13:40:04	CREATE TABLE Account (Account_No INT PRIMARY...	Error Code: 1050. Table 'account' already exists
14	13:40:04	CREATE TABLE Bank_Transaction (Transaction_ID IN...	Error Code: 1050. Table 'bank_transaction' already exists
15	13:40:04	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
16	13:41:25	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

Object Info Session

7. INSERT ACCOUNT DATA

The screenshot shows the SQL Developer interface with the following SQL script in the main editor:

```
41 REFERENCES Customer(Customer_ID)
42 );
43 INSERT INTO Customer
44 (Customer_ID, Customer_Name, Phone, Email, City)
45 VALUES
46 (101, 'Ravi Kumar', '9876543210', 'ravi@gmail.com', 'Hyderabad'),
47 (102, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'),
48 (103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),
49 (104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
50 (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');
51 INSERT INTO Account
52 (Account_No, Customer_ID, Account_Type, Balance, Branch)
53 VALUES
54 (10001, 101, 'Savings', 50000, 'Hyderabad'),
55 (10002, 102, 'Savings', 75000, 'Vijayawada'),
56 (10003, 103, 'Current', 120000, 'Bangalore'),
57 (10004, 104, 'Savings', 45000, 'Chennai'),
58 (10005, 105, 'Current', 90000, 'Hyderabad');
```

The Output window shows the following messages:

#	Time	Action	Message
11	13:40:04	USE BankDB	0 row(s) affected
12	13:40:04	CREATE TABLE Customer (Customer_ID INT PRIMAR...	Error Code: 1050. Table 'customer' already exists
13	13:40:04	CREATE TABLE Account (Account_No INT PRIMARY...	Error Code: 1050. Table 'account' already exists
14	13:40:04	CREATE TABLE Bank_Transaction (Transaction_ID IN...	Error Code: 1050. Table 'bank_transaction' already exists
15	13:40:04	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
16	13:41:25	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
17	13:42:34	INSERT INTO Account (Account_No, Customer_ID, Acco...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

8. INSERT TRANSACTION DATA

The screenshot shows the SQL Developer interface with the following SQL script in the main editor:

```
52 (Account_No, Customer_ID, Account_Type, Balance, Branch)
53 VALUES
54 (10001, 101, 'Savings', 50000, 'Hyderabad'),
55 (10002, 102, 'Savings', 75000, 'Vijayawada'),
56 (10003, 103, 'Current', 120000, 'Bangalore'),
57 (10004, 104, 'Savings', 45000, 'Chennai'),
58 (10005, 105, 'Current', 90000, 'Hyderabad');
59 INSERT INTO Bank_Transaction
60 (Account_No, Transaction_Type, Amount)
61 VALUES
62 (10001, 'DEPOSIT', 10000),
63 (10002, 'DEPOSIT', 15000),
64 (10003, 'WITHDRAW', 20000),
65 (10004, 'DEPOSIT', 5000),
66 (10005, 'WITHDRAW', 10000);
```

The Output window shows the following messages:

#	Time	Action	Message
9	13:39:08	CREATE TABLE Bank_Transaction (Transaction_ID IN...	0 row(s) affected
10	13:40:04	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; databa...
11	13:40:04	USE BankDB	0 row(s) affected
12	13:40:04	CREATE TABLE Customer (Customer_ID INT PRIMAR...	Error Code: 1050. Table 'customer' already exists
13	13:40:04	CREATE TABLE Account (Account_No INT PRIMARY...	Error Code: 1050. Table 'account' already exists
14	13:40:04	CREATE TABLE Bank_Transaction (Transaction_ID IN...	Error Code: 1050. Table 'bank_transaction' already exists
15	13:40:04	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
16	13:41:25	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
17	13:42:34	INSERT INTO Account (Account_No, Customer_ID, Acco...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
18	13:43:54	INSERT INTO Bank_Transaction (Account_No, Transacti...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

9. INSERT LOAN DATA

The screenshot shows the SQL Developer interface with a script titled "TRIGGERS_AND_STORED_P...". The script contains the following SQL statements:

```
59 INSERT INTO Bank_Transaction
60 (Account_No, Transaction_Type, Amount)
61 VALUES
62 (10001, 'DEPOSIT', 10000),
63 (10002, 'DEPOSIT', 15000),
64 (10003, 'WITHDRAW', 20000),
65 (10004, 'DEPOSIT', 5000),
66 (10005, 'WITHDRAW', 10000);
67 INSERT INTO Loan
68 (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
69 VALUES
70 (501, 101, 'Home Loan', 5000000, 7.5),
71 (502, 102, 'Education Loan', 1000000, 6.5),
72 (503, 103, 'Car Loan', 800000, 8.2),
73 (504, 104, 'Personal Loan', 500000, 10.5);
```

The Output window shows the execution results:

#	Time	Action	Message
10	13:40:04	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; databas...
11	13:40:04	USE BankDB	0 row(s) Error Code: 1007. Can't create database 'bank...
12	13:40:04	CREATE TABLE Customer (Customer_ID INT PRIMAR...	Error Code: 1050. Table 'customer' already exists
13	13:40:04	CREATE TABLE Account (Account_No INT PRIMARY...	Error Code: 1050. Table 'account' already exists
14	13:40:04	CREATE TABLE Bank_Transaction (Transaction_ID IN...	Error Code: 1050. Table 'bank_transaction' already exists
15	13:40:04	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
16	13:41:25	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
17	13:42:34	INSERT INTO Account (Account_No, Customer_ID, Acco...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
18	13:43:54	INSERT INTO Bank_Transaction (Account_No, Transacti...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
19	13:44:59	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, ...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0

10. DISPLAY TABLE DATA

The screenshot shows the SQL Developer interface with a script titled "TRIGGERS_AND_STORED_P...". The script contains the following SQL statements:

```
70 (501, 101, 'Home Loan', 5000000, 7.5),
71 (502, 102, 'Education Loan', 1000000, 6.5),
72 (503, 103, 'Car Loan', 800000, 8.2),
73 (504, 104, 'Personal Loan', 500000, 10.5);
74 SELECT * FROM Customer;
75 SELECT * FROM Account;
76 SELECT * FROM Bank_Transaction;
77 SELECT * FROM Loan;
```

The Output window shows the execution results:

#	Time	Action	Message
12	13:40:04	CREATE TABLE Customer (Customer_ID INT PRIMA...	Error Code: 1050. Table 'customer' already exists
13	13:40:04	CREATE TABLE Account (Account_No INT PRIMAR...	Error Code: 1050. Table 'account' already exists
14	13:40:04	CREATE TABLE Bank_Transaction (Transaction_ID I...	Error Code: 1050. Table 'bank_transaction' already exists
15	13:40:04	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
16	13:41:25	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
17	13:42:34	INSERT INTO Account (Account_No, Customer_ID, Acco...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
18	13:43:54	INSERT INTO Bank_Transaction (Account_No, Transact...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
19	13:44:59	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, ...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
20	13:46:03	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned

The Result Grid shows the data from the Customer table:

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

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Edit View Query Database Server Tools Scripting Help

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Limit to 1000 rows

SQLAdditions

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EMAS

Filter objects

bank_db
bank_managem
bankdb
Tables
Views
Stored Proce
Functions
bookflow_db
emr_portal
hospital_db
joins_db
practice_db
sakila
student_db
student_info
student_marks1
student_marks_c
students_db
sys
world

70 (501, 101, 'Home Loan', 5000000, 7.5),
71 (502, 102, 'Education Loan', 1000000, 6.5),
72 (503, 103, 'Car Loan', 800000, 8.2),
73 (504, 104, 'Personal Loan', 500000, 10.5);
74 • SELECT * FROM Customer;
75 • SELECT * FROM Account;
76 • SELECT * FROM Bank_Transaction;
77 • SELECT * FROM Loan;

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad

Account 2 x

Apply Revert Context Help

Output

Action Output

#	Time	Action	Message
13	13:40:04	CREATE TABLE Account (Account_No INT PRIMAR...	Error Code: 1050. Table 'account' already exists
14	13:40:04	CREATE TABLE Bank_Transaction (Transaction_ID I...	Error Code: 1050. Table 'bank_transaction' already exists
15	13:40:04	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
16	13:41:25	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
17	13:42:34	INSERT INTO Account (Account_No, Customer_ID, Acc...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
18	13:43:54	INSERT INTO Bank_Transaction (Account_No, Transact...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
19	13:44:59	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
20	13:46:03	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
21	13:47:41	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned

ect Info Session

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Edit View Query Database Server Tools Scripting Help

igator TRIGGERS_AND_STORED_P... x

Limit to 1000 rows

SQLAdditions

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EMAS

Filter objects

bank_db
bank_managem
bankdb
Tables
Views
Stored Proce
Functions
bookflow_db
emr_portal
hospital_db
joins_db
practice_db
sakila
student_db
student_info
student_marks1
student_marks_c
students_db
sys
world

70 (501, 101, 'Home Loan', 5000000, 7.5),
71 (502, 102, 'Education Loan', 1000000, 6.5),
72 (503, 103, 'Car Loan', 800000, 8.2),
73 (504, 104, 'Personal Loan', 500000, 10.5);
74 • SELECT * FROM Customer;
75 • SELECT * FROM Account;
76 • SELECT * FROM Bank_Transaction;
77 • SELECT * FROM Loan;

Result Grid

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-11 13:43:54
2	10002	DEPOSIT	15000.00	2026-08-11 13:43:54
3	10003	WITHDRAW	20000.00	2026-08-11 13:43:54
4	10004	DEPOSIT	5000.00	2026-08-11 13:43:54
5	10005	WITHDRAW	10000.00	2026-08-11 13:43:54

Bank_Transaction 3 x

Apply Revert Context Help

Output

Action Output

#	Time	Action	Message
14	13:40:04	CREATE TABLE Bank_Transaction (Transaction_ID I...	Error Code: 1050. Table 'bank_transaction' already exists
15	13:40:04	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
16	13:41:25	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
17	13:42:34	INSERT INTO Account (Account_No, Customer_ID, Acc...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
18	13:43:54	INSERT INTO Bank_Transaction (Account_No, Transact...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
19	13:44:59	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
20	13:46:03	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
21	13:47:41	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
22	13:48:10	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned

ject Info Session

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Edit View Query Database Server Tools Scripting Help

TRIGGERS_AND_STORED_P...

Limit to 1000 rows

SQLAdditions

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70 (501, 101, 'Home Loan', 5000000, 7.5),
 71 (502, 102, 'Education Loan', 1000000, 6.5),
 72 (503, 103, 'Car Loan', 800000, 8.2),
 73 (504, 104, 'Personal Loan', 500000, 10.5);
 74 • SELECT * FROM Customer;
 75 • SELECT * FROM Account;
 76 • SELECT * FROM Bank_Transaction;
 77 • SELECT * FROM Loan;

Result Grid

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
NULL	NULL	NULL	NULL	NULL

Loan 4 x Apply Revert Context Help

Output

Action Output

#	Time	Action	Message
15	13:40:04	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
16	13:41:25	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
17	13:42:34	INSERT INTO Account (Account_No, Customer_ID, Acc...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
18	13:43:54	INSERT INTO Bank_Transaction (Account_No, Transact...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
19	13:44:59	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type....	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
20	13:46:03	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
21	13:47:41	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
22	13:48:10	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
23	13:48:43	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned

ect Info Session

11. PROCEDURE TO DISPLAY ALL CUSTOMERS

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Edit View Query Database Server Tools Scripting Help

TRIGGERS_AND_STORED_P...

Limit to 1000 rows

SQLAdditions

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77 • SELECT * FROM Loan;
 78 DELIMITER //
 79 • CREATE PROCEDURE GetAllCustomers()
 80 BEGIN
 81 SELECT * FROM Customer;
 82 END //
 83 DELIMITER ;
 84 • CALL GetAllCustomers();

Result Grid

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543212	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

Customer 5 Account 6 Bank_Transaction 7 Loan 8 Result 9 x Read Only Context Help

Output

Action Output

#	Time	Action	Message
31	13:55:24	INSERT INTO Account (Account_No, Customer_ID, Acc...	Error Code: 1062. Duplicate entry '10001' for key 'account...
32	13:55:24	INSERT INTO Bank_Transaction (Account_No, Transact...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
33	13:55:24	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type....	Error Code: 1062. Duplicate entry '501' for key 'loan.PRIM...
34	13:55:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
35	13:55:24	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
36	13:55:24	SELECT * FROM Bank_Transaction LIMIT 0, 1000	10 row(s) returned
37	13:55:24	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned
38	13:55:24	CREATE PROCEDURE GetAllCustomers() BEGIN SEL...	0 row(s) affected
39	13:55:24	CALL GetAllCustomers()	5 row(s) returned

ject Info Session

12. PROCEDURE TO DISPLAY ACCOUNT DETAILS FOR A GIVEN ACCOUNT NUMBER

The screenshot shows the SQL Developer interface with the 'TRIGGERS_AND_STORED_P...' window open. The SQL script in the editor is as follows:

```
87      IN p_Account_No INT
88  )
89  BEGIN
90      SELECT *
91      FROM Account
92      WHERE Account_No = p_Account_No;
93  END //
94  DELIMITER ;
95  CALL GetAccountDetails(10001);
```

The 'Result Grid' shows the output of the procedure:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad

The 'Output' window shows the execution log:

#	Time	Action	Message
49	13:58:01	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type,...	Error Code: 1062. Duplicate entry '501' for key loan.PRIM...
50	13:58:01	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
51	13:58:01	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
52	13:58:01	SELECT * FROM Bank_Transaction LIMIT 0, 1000	15 row(s) returned
53	13:58:01	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned
54	13:58:01	CREATE PROCEDURE GetAllCustomers() BEGIN SEL...	Error Code: 1304. PROCEDURE GetAllCustomers already...
55	13:58:01	CALL GetAllCustomers()	5 row(s) returned
56	13:58:01	CREATE PROCEDURE GetAccountDetails(IN p_Acc...	0 row(s) affected
57	13:58:01	CALL GetAccountDetails(10001)	1 row(s) returned

13. PROCEDURE TO FIND CUSTOMER ACCOUNTS

The screenshot shows the SQL Developer interface with the 'TRIGGERS_AND_STORED_P...' window open. The SQL script in the editor is as follows:

```
107      A.Balance,
108      A.Branch
109  FROM Customer C
110  JOIN Account A
111  ON C.Customer_ID = A.Customer_ID
112  WHERE C.Customer_ID = p_Customer_ID;
113  END //
114  DELIMITER ;
115  CALL GetCustomerAccounts(101);
```

The 'Result Grid' shows the output of the procedure:

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	10001	Savings	50000.00	Hyderabad

The 'Output' window shows the execution log:

#	Time	Action	Message
12	08:29:52	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
13	08:29:52	SELECT * FROM Bank_Transaction LIMIT 0, 1000	20 row(s) returned
14	08:29:52	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned
15	08:29:52	CREATE PROCEDURE GetAllCustomers() BEGIN SEL...	Error Code: 1304. PROCEDURE GetAllCustomers already...
16	08:29:52	CALL GetAllCustomers()	5 row(s) returned
17	08:29:52	CREATE PROCEDURE GetAccountDetails(IN p_Acc...	Error Code: 1304. PROCEDURE GetAccountDetails alrea...
18	08:29:52	CALL GetAccountDetails(10001)	1 row(s) returned
19	08:29:52	CREATE PROCEDURE GetCustomerAccounts(IN p_...	0 row(s) affected
20	08:29:52	CALL GetCustomerAccounts(101)	1 row(s) returned

14. PROCEDURE TO DEPOSIT MONEY

The screenshot shows the SQL Developer interface with the 'TRIGGERS_AND_STORED_P...' window open. The SQL editor contains the following code:

```
121     IN p_Amount DECIMAL(12,2)
122 )
123 BEGIN
124     UPDATE Account
125     SET Balance = Balance + p_Amount
126     WHERE Account_No = p_Account_No;
127 END //
128 DELIMITER ;
129 CALL DepositMoney(10001, 5000);
```

The 'Result Grid' shows the following data:

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	10001	Savings	50000.00	Hyderabad

The 'Output' window shows the following messages:

#	Time	Action	Message
34	08:32:04	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned
35	08:32:04	CREATE PROCEDURE GetAllCustomers() BEGIN SEL...	Error Code: 1304. PROCEDURE GetAllCustomers already e...
36	08:32:04	CALL GetAllCustomers()	5 row(s) returned
37	08:32:04	CREATE PROCEDURE GetAccountDetails(IN p_Acc...	Error Code: 1304. PROCEDURE GetAccountDetails alrea...
38	08:32:04	CALL GetAccountDetails(10001)	1 row(s) returned
39	08:32:04	CREATE PROCEDURE GetCustomerAccounts(IN p_...	Error Code: 1304. PROCEDURE GetCustomerAccounts a...
40	08:32:04	CALL GetCustomerAccounts(101)	1 row(s) returned
41	08:32:04	CREATE PROCEDURE DepositMoney(IN p_Account...	0 row(s) affected
42	08:32:04	CALL DepositMoney(10001, 5000)	1 row(s) affected

15. PROCEDURE TO WITHDRAW MONEY

The screenshot shows the SQL Developer interface with the 'TRIGGERS_AND_STORED_P...' window open. The SQL editor contains the following code:

```
135     IN p_Amount DECIMAL(12,2)
136 )
137 BEGIN
138     UPDATE Account
139     SET Balance = Balance - p_Amount
140     WHERE Account_No = p_Account_No;
141 END //
142 DELIMITER ;
143 CALL WithdrawMoney(10001, 3000);
```

The 'Result Grid' shows the following data:

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	10001	Savings	55000.00	Hyderabad

The 'Output' window shows the following messages:

#	Time	Action	Message
58	08:54:43	CALL GetAllCustomers()	5 row(s) returned
59	08:54:43	CREATE PROCEDURE GetAccountDetails(IN p_Acc...	Error Code: 1304. PROCEDURE GetAccountDetails alrea...
60	08:54:43	CALL GetAccountDetails(10001)	1 row(s) returned
61	08:54:43	CREATE PROCEDURE GetCustomerAccounts(IN p_...	Error Code: 1304. PROCEDURE GetCustomerAccounts a...
62	08:54:43	CALL GetCustomerAccounts(101)	1 row(s) returned
63	08:54:43	CREATE PROCEDURE DepositMoney(IN p_Account...	Error Code: 1304. PROCEDURE DepositMoney already e...
64	08:54:43	CALL DepositMoney(10001, 5000)	1 row(s) affected
65	08:54:43	CREATE PROCEDURE WithdrawMoney(IN p_Accou...	0 row(s) affected
66	08:54:43	CALL WithdrawMoney(10001, 3000)	1 row(s) affected

16. TRIGGER – PREVENT NEGATIVE BALANCE

The screenshot shows the SQL Developer interface with a script editor containing the following SQL code:

```
151 SIGNAL SQLSTATE '45000'
152 SET MESSAGE_TEXT =
153 'Transaction failed: Insufficient balance';
154 END IF;
155 END //
156 DELIMITER ;
157 UPDATE Account
158 SET Balance = Balance - 60000
159 WHERE Account_No = 10001;
```

The **Result Grid** shows the following data:

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	10001	Savings	57000.00	Hyderabad

The **Output** pane shows the execution results:

#	Time	Action	Message
84	08:56:25	CALL GetAccountDetails(10001)	1 row(s) returned
85	08:56:25	CREATE PROCEDURE GetCustomerAccounts(IN p_...	Error Code: 1304. PROCEDURE GetCustomerAccounts a...
86	08:56:25	CALL GetCustomerAccounts(101)	1 row(s) returned
87	08:56:25	CREATE PROCEDURE DepositMoney(IN p_Account...	Error Code: 1304. PROCEDURE DepositMoney already e...
88	08:56:25	CALL DepositMoney(10001, 5000)	1 row(s) affected
89	08:56:25	CREATE PROCEDURE WithdrawMoney(IN p_Accou...	Error Code: 1304. PROCEDURE WithdrawMoney already...
90	08:56:25	CALL WithdrawMoney(10001, 3000)	1 row(s) affected
91	08:56:25	CREATE TRIGGER CheckBalance BEFORE UPDATE O...	0 row(s) affected
92	08:56:25	UPDATE Account SET Balance = Balance - 60000 WHE...	Error Code: 1644. Transaction failed: Insufficient balance

17. TRIGGER – PREVENT NEGATIVE DEPOSIT

The screenshot shows the SQL Developer interface with a script editor containing the following SQL code:

```
169 SET MESSAGE_TEXT =
170 'Transaction amount must be greater than zero';
171 END IF;
172 END //
173 DELIMITER ;
174 INSERT INTO Bank_Transaction
175 (Account_No, Transaction_Type, Amount)
176 VALUES
177 (10001, 'DEPOSIT', -5000);
```

The **Result Grid** shows the following data:

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	10001	Savings	59000.00	Hyderabad

The **Output** pane shows the execution results:

#	Time	Action	Message
112	09:03:56	CALL GetCustomerAccounts(101)	1 row(s) returned
113	09:03:56	CREATE PROCEDURE DepositMoney(IN p_Account...	Error Code: 1304. PROCEDURE DepositMoney already e...
114	09:03:56	CALL DepositMoney(10001, 5000)	1 row(s) affected
115	09:03:56	CREATE PROCEDURE WithdrawMoney(IN p_Accou...	Error Code: 1304. PROCEDURE WithdrawMoney already...
116	09:03:56	CALL WithdrawMoney(10001, 3000)	1 row(s) affected
117	09:03:56	CREATE TRIGGER CheckBalance BEFORE UPDATE O...	Error Code: 1359. Trigger already exists
118	09:03:56	UPDATE Account SET Balance = Balance - 60000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
119	09:03:56	CREATE TRIGGER CheckTransactionAmount BEFORE I...	0 row(s) affected
120	09:03:56	INSERT INTO Bank_Transaction (Account_No, Transact...	Error Code: 1644. Transaction amount must be greater th...

18. CREATE TRANSACTION AUDIT TABLE

Automatically disabled. To manually current ca toggle a

```

178
179 CREATE TABLE Transaction_Audit (
180     Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
181     Transaction_ID INT,
182     Account_No INT,
183     Transaction_Type VARCHAR(20),
184     Amount DECIMAL(12,2),
185     Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
186 );

```

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	10001	Savings	1000.00	Hyderabad

Output

#	Time	Action	Message
141	09:04:40	CREATE PROCEDURE DepositMoney(IN p_Account...	Error Code: 1304. PROCEDURE DepositMoney already e...
142	09:04:40	CALL DepositMoney(10001, 5000)	1 row(s) affected
143	09:04:40	CREATE PROCEDURE WithdrawMoney(IN p_Accou...	Error Code: 1304. PROCEDURE WithdrawMoney already...
144	09:04:40	CALL WithdrawMoney(10001, 3000)	1 row(s) affected
145	09:04:40	CREATE TRIGGER CheckBalance BEFORE UPDATE O...	Error Code: 1359. Trigger already exists
146	09:04:40	UPDATE Account SET Balance = Balance - 60000 WHE...	Error Code: 1644. Transaction failed: Insufficient balance
147	09:04:40	CREATE TRIGGER CheckTransactionAmount BEFORE I...	Error Code: 1359. Trigger already exists
148	09:04:40	INSERT INTO Bank_Transaction (Account_No, Transact...	Error Code: 1644. Transaction amount must be greater th...
149	09:04:40	CREATE TABLE Transaction_Audit (Audit_ID INT PRI...	0 row(s) affected

19. TRIGGER – TRANSACTION AUDIT

Automatically disabled. To manually current ca toggle a

```

206 NEW.Amount
207 );
208 END //
209 DELIMITER ;
210 INSERT INTO Bank_Transaction
211 (Account_No, Transaction_Type, Amount)
212 VALUES
213 (10001, 'DEPOSIT', 2500);
214 SELECT * FROM Transaction_Audit;

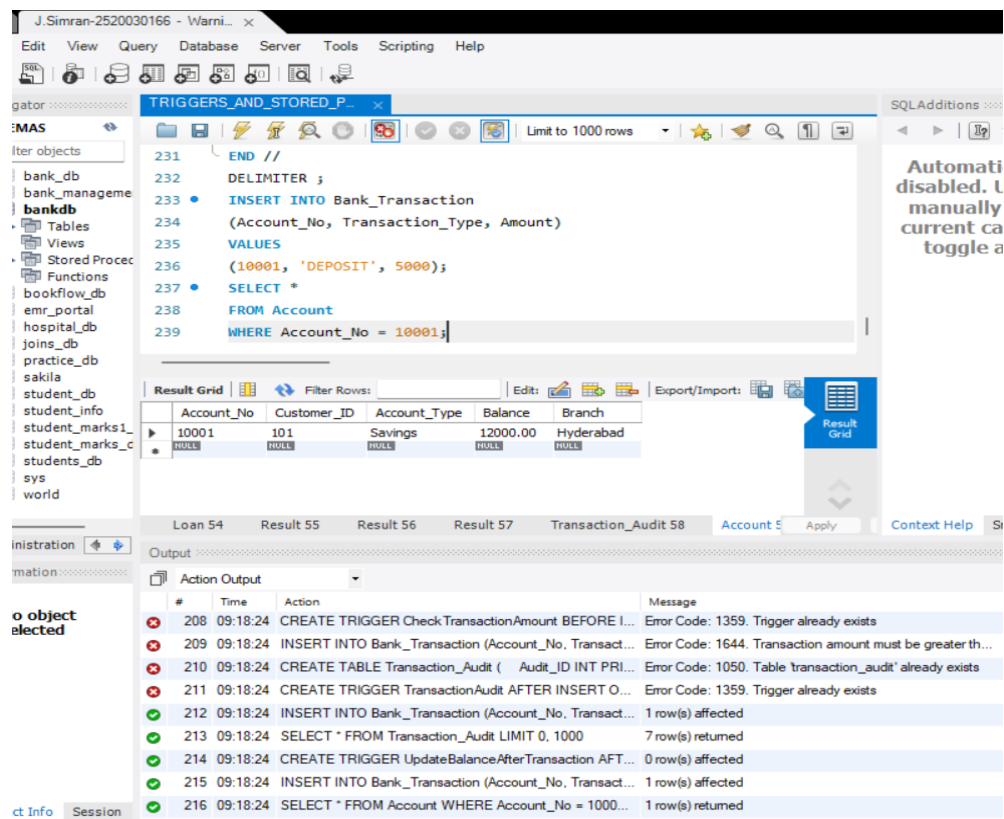
```

Audit_ID	Transaction_ID	Account_No	Transaction_Type	Amount	Audit_Date
1	51	10001	DEPOSIT	2500.00	2026-08-14 09:06:00

Output

#	Time	Action	Message
173	09:06:00	CALL WithdrawMoney(10001, 3000)	1 row(s) affected
174	09:06:00	CREATE TRIGGER CheckBalance BEFORE UPDATE O...	Error Code: 1359. Trigger already exists
175	09:06:00	UPDATE Account SET Balance = Balance - 60000 WHE...	Error Code: 1644. Transaction failed: Insufficient balance
176	09:06:00	CREATE TRIGGER CheckTransactionAmount BEFORE I...	Error Code: 1359. Trigger already exists
177	09:06:00	INSERT INTO Bank_Transaction (Account_No, Transact...	Error Code: 1644. Transaction amount must be greater th...
178	09:06:00	CREATE TABLE Transaction_Audit (Audit_ID INT PRI...	Error Code: 1050. Table transaction_audit' already exists
179	09:06:00	CREATE TRIGGER TransactionAudit AFTER INSERT O...	0 row(s) affected
180	09:06:00	INSERT INTO Bank_Transaction (Account_No, Transact...	1 row(s) affected
181	09:06:00	SELECT * FROM Transaction_Audit LIMIT 0, 1000	1 row(s) returned

20. TRIGGER – AUTOMATICALLY UPDATE ACCOUNT BALANCE



SQL Developer interface showing the creation of a trigger to automatically update account balance. The trigger is named 'CheckTransactionAmount' and is created before the 'Bank_Transaction' table. The trigger body inserts a row into 'Transaction_Audit' and updates the 'Balance' in 'Account'.

```

231 END //
232 DELIMITER ;
233 INSERT INTO Bank_Transaction
234 (Account_No, Transaction_Type, Amount)
235 VALUES
236 (10001, 'DEPOSIT', 5000);
237 SELECT *
238 FROM Account
239 WHERE Account_No = 10001;

```

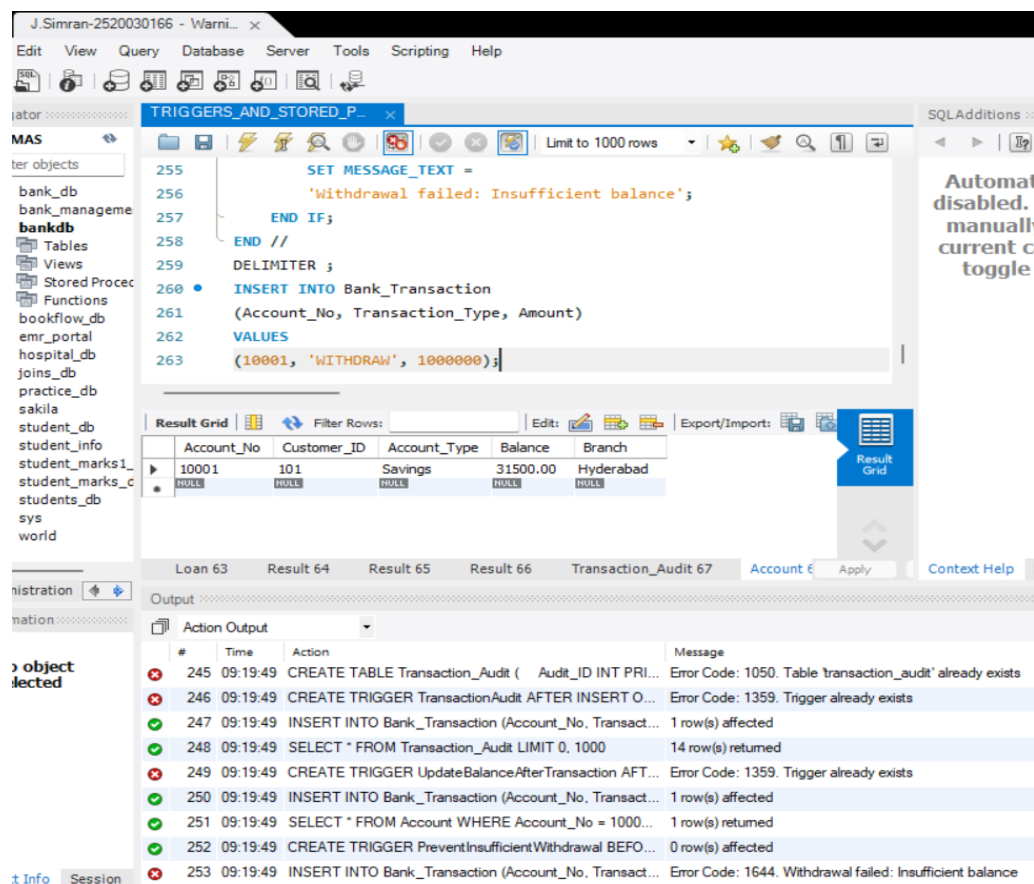
Result Grid:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	12000.00	Hyderabad

Action Output:

#	Time	Action	Message
208	09:18:24	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction	Error Code: 1359. Trigger already exists
209	09:18:24	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 5000);	Error Code: 1644. Transaction amount must be greater than 0
210	09:18:24	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY, Account_No INT, Transaction_Type VARCHAR(50), Amount DECIMAL(10,2), Transaction_Time TIMESTAMP)	Error Code: 1050. Table 'transaction_audit' already exists
211	09:18:24	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction	Error Code: 1359. Trigger already exists
212	09:18:24	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 5000);	1 row(s) affected
213	09:18:24	SELECT * FROM Transaction_Audit LIMIT 0, 1000	7 row(s) returned
214	09:18:24	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER INSERT ON Transaction_Audit	0 row(s) affected
215	09:18:24	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 5000);	1 row(s) affected
216	09:18:24	SELECT * FROM Account WHERE Account_No = 10001;	1 row(s) returned

21. TRIGGER – PREVENT WITHDRAWAL ABOVE BALANCE



SQL Developer interface showing the creation of a trigger to prevent withdrawal above balance. The trigger is named 'PreventInsufficientWithdrawal' and is created before the 'Bank_Transaction' table. The trigger body inserts a row into 'Transaction_Audit' and updates the 'Balance' in 'Account'.

```

255 SET MESSAGE_TEXT =
256 'Withdrawal failed: Insufficient balance';
257 END IF;
258 END //
259 DELIMITER ;
260 INSERT INTO Bank_Transaction
261 (Account_No, Transaction_Type, Amount)
262 VALUES
263 (10001, 'WITHDRAW', 1000000);

```

Result Grid:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	31500.00	Hyderabad

Action Output:

#	Time	Action	Message
245	09:19:49	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY, Account_No INT, Transaction_Type VARCHAR(50), Amount DECIMAL(10,2), Transaction_Time TIMESTAMP)	Error Code: 1050. Table 'transaction_audit' already exists
246	09:19:49	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction	Error Code: 1359. Trigger already exists
247	09:19:49	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'WITHDRAW', 1000000);	1 row(s) affected
248	09:19:49	SELECT * FROM Transaction_Audit LIMIT 0, 1000	14 row(s) returned
249	09:19:49	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER INSERT ON Transaction_Audit	Error Code: 1359. Trigger already exists
250	09:19:49	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'WITHDRAW', 1000000);	1 row(s) affected
251	09:19:49	SELECT * FROM Account WHERE Account_No = 10001;	1 row(s) returned
252	09:19:49	CREATE TRIGGER PreventInsufficientWithdrawal BEFORE INSERT ON Bank_Transaction	0 row(s) affected
253	09:19:49	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'WITHDRAW', 1000000);	Error Code: 1644. Withdrawal failed: Insufficient balance

22. PROCEDURE – ACCOUNT TRANSFER

The screenshot shows the SQL Developer interface with the 'TRIGGERS_AND_STORED_P...' window open. The SQL editor contains the following code:

```
287 SET Balance = Balance + TransferAmount
288 WHERE Account_No = ReceiverAccount;
289 END IF;
290 END //
291 DELIMITER ;
292 CALL TransferMoney(10001, 10002, 5000);
293 SELECT *
294 FROM Account
295 WHERE Account_No IN (10001,10002);
```

The 'Result Grid' shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	46000.00	Hyderabad
10002	102	Savings	110000.00	Vijayawada

The 'Output' window shows the 'Action Output' for the execution of the procedure. The output is as follows:

#	Time	Action	Message
285	09:21:29	SELECT * FROM Transaction_Audit LIMIT 0, 1000	21 row(s) returned
286	09:21:29	CREATE TRIGGER UpdateBalanceAfterTransaction AFT...	Error Code: 1359. Trigger already exists
287	09:21:29	INSERT INTO Bank_Transaction (Account_No, Transact...	1 row(s) affected
288	09:21:29	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
289	09:21:29	CREATE TRIGGER PreventInsufficientWithdrawal BEFO...	Error Code: 1359. Trigger already exists
290	09:21:29	INSERT INTO Bank_Transaction (Account_No, Transact...	Error Code: 1644. Withdrawal failed: Insufficient balance
291	09:21:29	CREATE PROCEDURE TransferMoney(IN SenderAcc...	0 row(s) affected
292	09:21:29	CALL TransferMoney(10001, 10002, 5000)	1 row(s) affected
293	09:21:29	SELECT * FROM Account WHERE Account_No IN (100...	2 row(s) returned

23. PROCEDURE – CUSTOMER LOAN DETAILS

The screenshot shows the SQL Developer interface with the 'TRIGGERS_AND_STORED_P...' window open. The SQL editor contains the following code:

```
307 L.Loan_Amount,
308 L.Interest_Rate
309 FROM Customer C
310 JOIN Loan L
311 ON C.Customer_ID = L.Customer_ID
312 WHERE C.Customer_ID = p_Customer_ID;
313 END //
314 DELIMITER ;
315 CALL GetCustomerLoans(101);
```

The 'Result Grid' shows the following data:

Customer_Name	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
Ravi Kumar	501	Home Loan	5000000.00	7.50

The 'Output' window shows the 'Action Output' for the execution of the procedure. The output is as follows:

#	Time	Action	Message
327	09:28:37	INSERT INTO Bank_Transaction (Account_No, Transact...	1 row(s) affected
328	09:28:37	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
329	09:28:37	CREATE TRIGGER PreventInsufficientWithdrawal BEFO...	Error Code: 1359. Trigger already exists
330	09:28:37	INSERT INTO Bank_Transaction (Account_No, Transact...	Error Code: 1644. Withdrawal failed: Insufficient balance
331	09:28:37	CREATE PROCEDURE TransferMoney(IN SenderAcc...	Error Code: 1304. PROCEDURE TransferMoney already ...
332	09:28:37	CALL TransferMoney(10001, 10002, 5000)	1 row(s) affected
333	09:28:37	SELECT * FROM Account WHERE Account_No IN (100...	2 row(s) returned
334	09:28:37	CREATE PROCEDURE GetCustomerLoans(IN p_Cust...	0 row(s) affected
335	09:28:37	CALL GetCustomerLoans(101)	1 row(s) returned

24. PROCEDURE – HIGH BALANCE ACCOUNTS

The screenshot shows the SQL Developer interface with the 'TRIGGERS_AND_STORED_P...' window open. The SQL script in the editor is as follows:

```
321 )
322 BEGIN
323     SELECT *
324     FROM Account
325     WHERE Balance >= MinimumBalance
326     ORDER BY Balance DESC;
327 END //
328 DELIMITER ;
329 CALL HighBalanceAccounts(50000);
```

The 'Result Grid' shows the output of the query:

Account_No	Customer_ID	Account_Type	Balance	Branch
10002	102	Savings	150000.00	Vijayawada
10004	104	Savings	65000.00	Chennai
10005	105	Current	50000.00	Hyderabad

The 'Output' window shows the execution log:

#	Time	Action	Message
371	09:29:43	CREATE TRIGGER PreventInsufficientWithdrawal BEFO...	Error Code: 1359. Trigger already exists
372	09:29:43	INSERT INTO Bank_Transaction (Account_No, Transact...	Error Code: 1644. Withdrawal failed: Insufficient balance
373	09:29:43	CREATE PROCEDURE TransferMoney(IN SenderAcc...	Error Code: 1304. PROCEDURE TransferMoney already ...
374	09:29:43	CALL TransferMoney(10001, 10002, 5000)	1 row(s) affected
375	09:29:43	SELECT * FROM Account WHERE Account_No IN (100...	2 row(s) returned
376	09:29:43	CREATE PROCEDURE GetCustomerLoans(IN p_Cust...	Error Code: 1304. PROCEDURE GetCustomerLoans alrea...
377	09:29:43	CALL GetCustomerLoans(101)	1 row(s) returned
378	09:29:43	CREATE PROCEDURE HighBalanceAccounts(IN Min...	0 row(s) affected
379	09:29:43	CALL HighBalanceAccounts(50000)	3 row(s) returned

25. PROCEDURE WITH IN, OUT PARAMETERS

The screenshot shows the SQL Developer interface with the 'TRIGGERS_AND_STORED_P...' window open. The SQL script in the editor is as follows:

```
336 BEGIN
337     SELECT Balance
338     INTO p_Balance
339     FROM Account
340     WHERE Account_No = p_Account_No;
341 END //
342 DELIMITER ;
343 CALL GetBalance(10001, @CurrentBalance);
344 SELECT @CurrentBalance;
```

The 'Result Grid' shows the output of the query:

@CurrentBalance
29500.00

The 'Output' window shows the execution log:

#	Time	Action	Message
418	09:31:18	CALL TransferMoney(10001, 10002, 5000)	1 row(s) affected
419	09:31:18	SELECT * FROM Account WHERE Account_No IN (100...	2 row(s) returned
420	09:31:18	CREATE PROCEDURE GetCustomerLoans(IN p_Cust...	Error Code: 1304. PROCEDURE GetCustomerLoans alrea...
421	09:31:18	CALL GetCustomerLoans(101)	1 row(s) returned
422	09:31:18	CREATE PROCEDURE HighBalanceAccounts(IN Min...	Error Code: 1304. PROCEDURE HighBalanceAccounts a...
423	09:31:18	CALL HighBalanceAccounts(50000)	2 row(s) returned
424	09:31:18	CREATE PROCEDURE GetBalance(IN p_Account_N...	0 row(s) affected
425	09:31:18	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected
426	09:31:18	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned

26. TRIGGER MANAGEMENT COMMANDS

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TRIGGERS_AND_STORED_P...

Limit to 1000 rows

```
340 WHERE Account_No = p_Account_No;
341 END //
342 DELIMITER ;
343 CALL GetBalance(10001, @CurrentBalance);
344 SELECT @CurrentBalance;
345 SHOW TRIGGERS;
346
```

Result Grid

Trigger	Event	Table	Statement
CheckBalance	UPDATE	account	BEGIN IF NEW.Balance < 0 THEN
CheckTransactionAmount	INSERT	bank_transaction	BEGIN IF NEW.Amount <= 0 THEN
PreventInsufficientWithdrawal	INSERT	bank_transaction	BEGIN DECLARE CurrentBalance C
TransactionAudit	INSERT	bank_transaction	BEGIN INSERT INTO Transaction_C
UpdateBalanceAfterTransaction	INSERT	bank_transaction	BEGIN IF NEW.Transaction_Type

Output

Action Output

#	Time	Action	Message
✓ 466	09:32:15	SELECT * FROM Account WHERE Account_No IN (100...	2 row(s) returned
✗ 467	09:32:15	CREATE PROCEDURE GetCustomerLoans(IN p_Cust...	Error Code: 1304. PROCEDURE GetCustomerLoans alrea...
✗ 468	09:32:15	CALL GetCustomerLoans(101)	1 row(s) returned
✗ 469	09:32:15	CREATE PROCEDURE HighBalanceAccounts(IN Min...	Error Code: 1304. PROCEDURE HighBalanceAccounts a...
✓ 470	09:32:15	CALL HighBalanceAccounts(50000)	2 row(s) returned
✗ 471	09:32:15	CREATE PROCEDURE GetBalance(IN p_Account_N...	Error Code: 1304. PROCEDURE GetBalance already exists
✓ 472	09:32:15	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected
✓ 473	09:32:15	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned
✓ 474	09:32:15	SHOW TRIGGERS	5 row(s) returned

Session

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Edit View Query Database Server Tools Scripting Help

TRIGGERS_AND_STORED_P...

Limit to 1000 rows

```
340 WHERE Account_No = p_Account_No;
341 END //
342 DELIMITER ;
343 CALL GetBalance(10001, @CurrentBalance);
344 SELECT @CurrentBalance;
345 SHOW TRIGGERS;
346 SHOW CREATE TRIGGER CheckBalance;
```

Result Grid

Trigger	sql_mode	SQL Original Statement
CheckBalance	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...	CREATE DEFINER='root'@'localh...

Output

Action Output

#	Time	Action	Message
✗ 515	09:32:51	CREATE PROCEDURE GetCustomerLoans(IN p_Cust...	Error Code: 1304. PROCEDURE GetCustomerLoans alrea...
✓ 516	09:32:51	CALL GetCustomerLoans(101)	1 row(s) returned
✗ 517	09:32:51	CREATE PROCEDURE HighBalanceAccounts(IN Min...	Error Code: 1304. PROCEDURE HighBalanceAccounts a...
✓ 518	09:32:51	CALL HighBalanceAccounts(50000)	2 row(s) returned
✗ 519	09:32:51	CREATE PROCEDURE GetBalance(IN p_Account_N...	Error Code: 1304. PROCEDURE GetBalance already exists
✓ 520	09:32:51	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected
✓ 521	09:32:51	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned
✓ 522	09:32:51	SHOW TRIGGERS	5 row(s) returned
✓ 523	09:32:51	SHOW CREATE TRIGGER CheckBalance	1 row(s) returned

Session

J.Simran-2520030166 - Warni... x

Edit View Query Database Server Tools Scripting Help

gator: TRIGGERS_AND_STORED_P_...

SQLAdditions

Automati disabled. U manually current ca toggle a

341 END //

342 DELIMITER ;

343 CALL GetBalance(10001, @CurrentBalance);

344 SELECT @CurrentBalance;

345 SHOW TRIGGERS;

346 SHOW CREATE TRIGGER CheckBalance;

347 DROP TRIGGER CheckBalance;

Result Grid Filter Rows: Export: Wrap Cell Content: I

Trigger	sql_mode	SQL Original Statement
CheckBalance	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...	CREATE DEFINER='root'@'localh

52 Account 153 Result 154 Result 155 Result 156 Result 157 Result 158 Read Only Context Help

Output

Action Output

#	Time	Action	Message
✓ 566	09:33:32	CALL GetCustomerLoans(101)	1 row(s) returned
✗ 567	09:33:32	CREATE PROCEDURE HighBalanceAccounts(IN Min...	Error Code: 1304. PROCEDURE HighBalanceAccounts a...
✓ 568	09:33:32	CALL HighBalanceAccounts(50000)	3 row(s) returned
✗ 569	09:33:32	CREATE PROCEDURE GetBalance(IN p_Account_N...	Error Code: 1304. PROCEDURE GetBalance already exists
✓ 570	09:33:32	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected
✓ 571	09:33:32	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned
✓ 572	09:33:32	SHOW TRIGGERS	5 row(s) returned
✓ 573	09:33:32	SHOW CREATE TRIGGER CheckBalance	1 row(s) returned
✓ 574	09:33:32	DROP TRIGGER CheckBalance	0 row(s) affected

ct Info Session

27. PROCEDURE MANAGEMENT COMMANDS

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SQLAdditions

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346 SHOW CREATE TRIGGER CheckBalance;

347 DROP TRIGGER CheckBalance;

348 SHOW PROCEDURE STATUS

349 WHERE Db = 'BankDB';

350 DROP PROCEDURE GetAllCustomers;

Result Grid Filter Rows: Export: Wrap Cell Content: I

Db	Name	Type	Language	Definer	Modified
bankdb	DepositMoney	PROCEDURE	SQL	root@localhost	2026-08-14 08:
bankdb	GetAccountDetails	PROCEDURE	SQL	root@localhost	2026-08-11 13:
bankdb	GetAllCustomers	PROCEDURE	SQL	root@localhost	2026-08-11 13:
bankdb	GetBalance	PROCEDURE	SQL	root@localhost	2026-08-14 09:
bankdb	GetCustomerAccounts	PROCEDURE	SQL	root@localhost	2026-08-14 08:
bankdb	GetCustomerLoans	PROCEDURE	SQL	root@localhost	2026-08-14 09:
bankdb	HighBalanceAccounts	PROCEDURE	SQL	root@localhost	2026-08-14 09:
bankdb	TransferMoney	PROCEDURE	SQL	root@localhost	2026-08-14 09:
bankdb	WithdrawMoney	PROCEDURE	SQL	root@localhost	2026-08-14 08:

168 Result 169 Result 170 Result 171 Result 172 Result 173 Result 174 Read Only Context Help

Output

Action Output

#	Time	Action	Message
✓ 618	09:34:42	CALL HighBalanceAccounts(50000)	3 row(s) returned
✗ 619	09:34:42	CREATE PROCEDURE GetBalance(IN p_Account_N...	Error Code: 1304. PROCEDURE GetBalance already exists
✓ 620	09:34:42	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected
✓ 621	09:34:42	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned
✓ 622	09:34:42	SHOW TRIGGERS	5 row(s) returned
✓ 623	09:34:42	SHOW CREATE TRIGGER CheckBalance	1 row(s) returned
✓ 624	09:34:42	DROP TRIGGER CheckBalance	0 row(s) affected
✓ 625	09:34:42	SHOW PROCEDURE STATUS WHERE Db = 'BankDB'	9 row(s) returned
✓ 626	09:34:42	DROP PROCEDURE GetAllCustomers	0 row(s) affected

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